

Assignment 3: Bridge Pattern

Name: Mukhammedali Bolegen

Group: SE-2538

Topic: Drawing

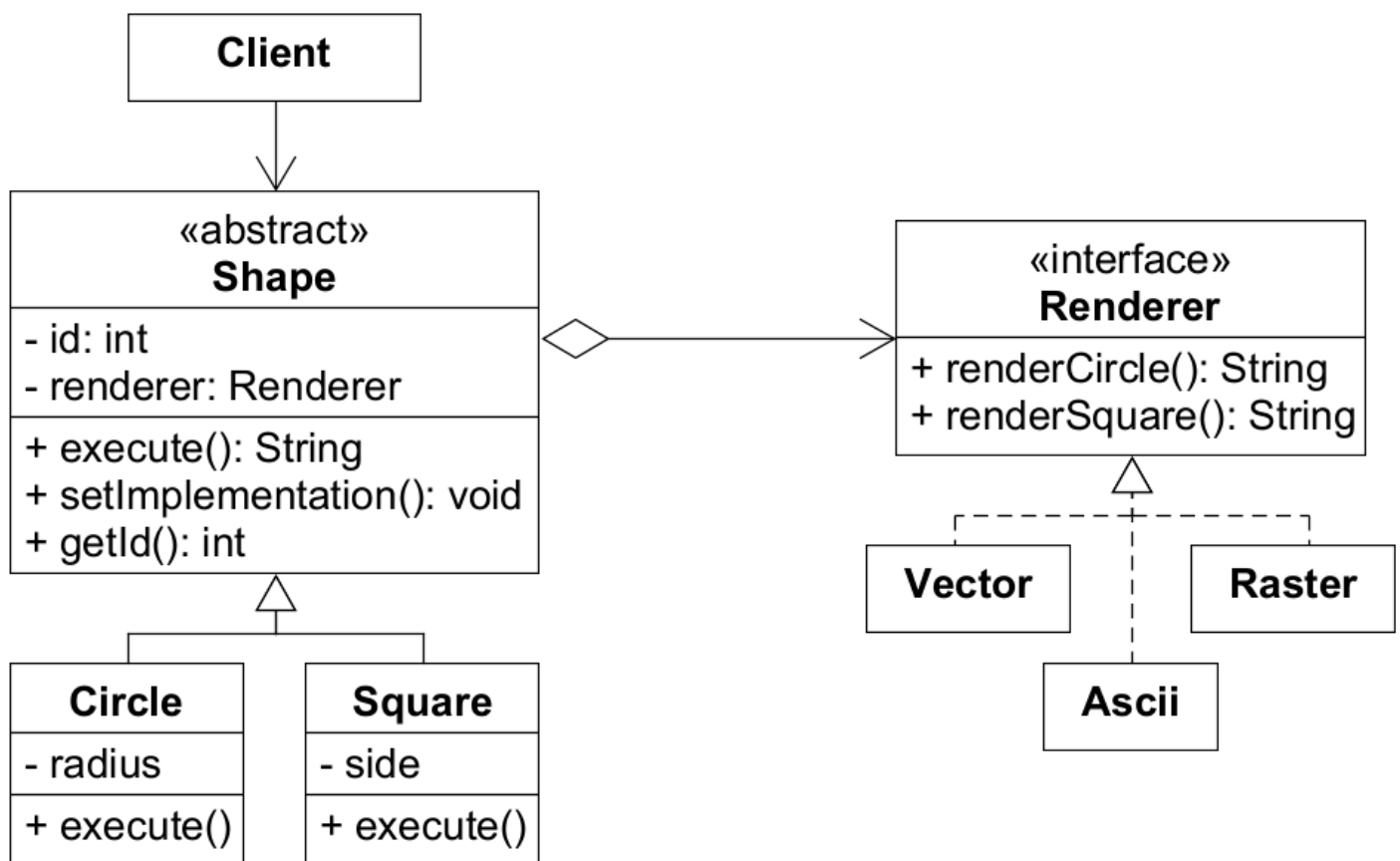
Dimensions and Pattern use

Two dimensions in this assignment are:

- Shapes (Circle, Square)
- Renderer (Vector, Raster, Ascii)

Each of which are independent of each other. If we were to use any other design pattern, we had to create one class for each combination of 2 dimensions (**N*M**), where **N** is amount of different shapes and **M** is amount of different renderers. But with Bridge pattern, we need to only create (**N+M**) classes (for each shape and renderer separately).

UML Diagram



Clean code principles

- **Separation of responsibilities:** Circle stores the radius and delegates rendering to Renderer; VectorRenderer, RasterRenderer, and AsciiRenderer handle rendering only.
- **Meaningful names:** Names like Circle, Square, Renderer, renderCircle(), and setImplementation() describe their purpose.
- **Small focused methods:** Circle.execute() only calls renderer.renderCircle(radius), while VectorRenderer.renderCircle() only creates the vector rendering result.
- **Avoidance of duplication:** One Circle can use any Renderer, so separate classes such as VectorCircle and RasterCircle are unnecessary. AsciiRenderer was added without duplicating the abstraction classes.
- **Encapsulation of state:** Shape keeps the ID, while Circle and Square keep their sizes in private fields. The renderer is changed through setImplementation() rather than directly by the client.